Tarun Kumar

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Education

- **Dr B. R. Ambedkar National Institute of Technology, Jalandhar** *August 2023 - Present*
  Bachelor of Technology

Experience

- **Data Analytics Intern - Edgenius Skillfied Mentor Pvt. Ltd.** (certificate) *Dec 2025 - Mar 2026*
  Stack: Python, Pandas (Data Cleaning), EDA, Statistics, Scikit-learn
  - Architected & deployed end-to-end data pipelines, encompassing data ingestion, ETL processing, and EDA on a dataset exceeding 50,000+ records, reducing null values from 18% to under 1% through systematic data cleaning and preprocessing.
  - Developed and delivered interactive BI dashboards and analytical reports using Tableau, translating complex dataset into actionable intelligence and enabling data-driven decision-making for cross-functional stakeholders.
  - Conducted statistical analysis and regression analysis to identify patterns, perform customer segmentation, and surface actionable business insights aligned with organizational KPIs and strategic objectives.
  - Collaborated cross-functionally with business units to align KPI definitions, standardize data governance practices, and ensure reporting accuracy.
  - Built and evaluated supervised machine learning models (including linear regression) using Scikit-learn and conducted feature importance analysis to forecast key business metrics and improve interpretability.
  - Engineered high-quality model inputs via feature engineering techniques, including feature scaling, categorical encoding, outlier detection, and normalization to improve predictive model accuracy and generalizability.

Skills

- **Analytics**: Exploratory Data Analysis (EDA), ETL, Statistical Analysis, Dimensional Data Modelling, Data Cleaning, KPI Design, Data Warehousing
- **Machine Learning:** Fundamentals of Machine Learning, Regression (Simple, Multiple & Polynomial Linear Regression), Scikit-learn, Feature Engineering, Statsmodels
- **Programming:** Python (Pandas, NumPy)
- **Statistics & Experiment Design:** Hypothesis Testing (Pearson/Spearman, Shapiro-Wilk, Kruskal-Wallis), OLS, Scipy, Model Evaluation (Precision, Recall, F1, AUC-ROC, Confusion Matrix)
- **Databases & SQL:** Fundamentals of Databases, MySQL Workbench, SQL(Group By, Subqueries, Joins, Window Functions)
- **Visualization & BI Tools:** Matplotlib, Seaborn, BI Tools (Tableau, Power BI), Excel, PowerPoint
- **Tools:** Jupyter Notebook, Git & GitHub, Streamlit, VS Code, PyCharm, Excel
- **BI and Reporting:** Dashboard Development, Data Modelling, Structured Problem Solving, Data Storytelling

Projects

- **Pneumonia Detection from Chest X-Rays (CNN, Transfer Learning):** *Python, TensorFlow/Keras, MobileNetV2, OpenCV, Scikit-learn*
  - Trained a MobileNetV2 transfer-learning CNN to classify chest X-rays as NORMAL / PNEUMONIA, achieving 91.2% accuracy, 93.3% precision, 92.6% recall, and 0.97 AUC-ROC on 624 held-out test images.
  - Handled ~3:2 class imbalance via computed class weights and augmentation (rotation, zoom, shift, flip); used early stopping and LR scheduling on val_auc to prevent overfitting on a ~5,800-image dataset.
  - Evaluated with precision, recall, F1, AUC-ROC, and a confusion matrix rather than accuracy alone, prioritizing recall on the PNEUMONIA class to minimize diagnostically costly false negatives.
  - Built and deployed a Streamlit app for real-time X-ray classification with confidence scores; published to GitHub.
- **Comprehensive EDA and Inferential Analysis(Cancer Data):** *Python, Pandas, Seaborn, Statistics, Scipy, Scikit-learn*
  - Performed exploratory and inferential statistical analysis in Python (pandas, seaborn) on a 50,000-record global cancer patient dataset spanning 10 countries and 8 cancer types. *(link)*
  - Applied hypothesis testing (Pearson/Spearman correlation, Shapiro-Wilk, Kruskal-Wallis) using SciPy to evaluate relationships between treatment cost, cancer stage, and survival years.
  - Built an OLS interaction model (statsmodels) to test whether genetic risk amplifies the effect of smoking on cancer severity.
  - Trained a Random Forest regression model (scikit-learn) to predict cancer severity scores, and ranked feature importance to identify smoking and genetic risk as top predictors.
  - Analyzed geographic disparities in treatment cost across countries and age groups, surfacing global healthcare affordability gaps.
- **End-to-End SQL Data Pipeline(Cleaning & EDA using SQL on Laptop Data):** *SQL, MySQL (link)*
  - Built an end-to-end SQL data pipeline (MySQL), includes cleaning, feature engineering, and EDA on a 1,300+ row laptop specifications dataset with zero external scripting.
  - Re-engineered table schema: fixed data types across 6+ columns, removed 60 duplicate records via groupby logic, and decomposed 4 compound columns (Cpu, Gpu, Memory, Resolution) into 11 structured fields using substring_index() and regexp_substr().
  - Performed native-SQL EDA using window functions (row_number(), count() over()) for percentile/quartile analysis and hand-implemented Pearson correlation via aggregate functions.
  - Applied IQR-based outlier detection, engineered new features (pixel-density/ppi), and one-hot encoded categorical data (40 columns) for ML-readiness.

Certificates

- **Master Data Analysis: Udemy (2025)** | Skills: Python & Its libraries, Statistics, SQL(MySQL), Tableau, Excel, PowerBI
- **GenAI Powered Data Analytics - Tata iQ (Forage), 2026**: EDA, Predictive Analytics, Process Automation using GenAI tools

Awards And Additional

- **Bronze medal**: 4×100 m relay, NIT Jalandhar Athletics Fest (Year 1)
- **National Cadet Corps (NCC)**: 21 PB BN NCC Kapurthala, (Jul 2024–Aug 2025), Gained Leadership and Discipline Skills
- **Hostel Discipline Committee Member**: Conflict Resolution, Discipline Monitoring, Team Coordination